Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach loss as an operation with a question and a unit.

Supply complete evaluation predictions and a training-derived baseline; retain reversed rankings and all large misses.

Keep one habit: Require per-record evidence and a declared metric before accepting a model-improvement claim.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Calculate MAE MSE and **RMSE**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on absolute and squared **loss**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Opposite signed residuals can cancel. Take each absolute value first, then add the magnitudes and divide by the count. The resulting MAE retains the target unit.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on **training** targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

End goal

By the end, I can explain **absolute error** and mae using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Mean, Median, and Baselines

Error Analysis

Confusion Matrix

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Concept to explain

Absolute error measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

Term	Meaning
absolute error	absolute value of actual minus predicted
MAE	mean of absolute errors
unit	measurement scale retained by MAE
cancellation	positive and negative errors offset

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the operation before the metric label

> Keep signed error,absolute magnitude,squared magnitude and root in distinct steps.

Main definitions and supplied calculation

A **loss** summarizes error magnitude without positive and negative **cancellation**. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly.

RMSE takes the square root of MSE so it returns to the target **unit**. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, **unit**s, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines absolute and squared **loss**.

Residuals [-1, 2,-3] cm have **absolute error**s [1, 2, 3], so MAE=6/3=2 cm. Squared errors [1, 4, 9] sum to 14, so MSE=14/3 square cm and **RMSE**=sqrt(14/3), about 2.16 cm. Keep the square and the average in the correct order.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Term	Meaning
MAE	Mean absolute error
MSE	Mean squared error
RMSE	Square root of mean squared error
Loss	A numerical penalty for prediction errors

Absolute and squared loss at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Residuals	minus 1 plus 2 minus 3	Signed differences
Absolute values	1 2 3	MAE 2 cm
Squares	1 4 9	MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Give MSE and RMSE with units. Explain the steps or evidence behind your response.

Complete candidate/baseline ranking contrast

Fit the mean baseline on separate training heights [5,7,9] cm, giving constant 7 cm; those targets are not evaluation cases. Candidate predictions [10,10,4] are supplied, not fitted in this lesson. On matched E1–E3, candidate residuals [0,0,6] give MAE 2 cm, MSE 12 cm², RMSE sqrt(12)≈3.46 cm. Baseline residuals [3,3,3] give MAE 3 cm, MSE 9 cm², RMSE 3 cm. The candidate is better by MAE but worse by MSE/RMSE here.

Explain the single large miss and state which loss was chosen for the task before scoring.

Read separate pictured residuals and unit scope

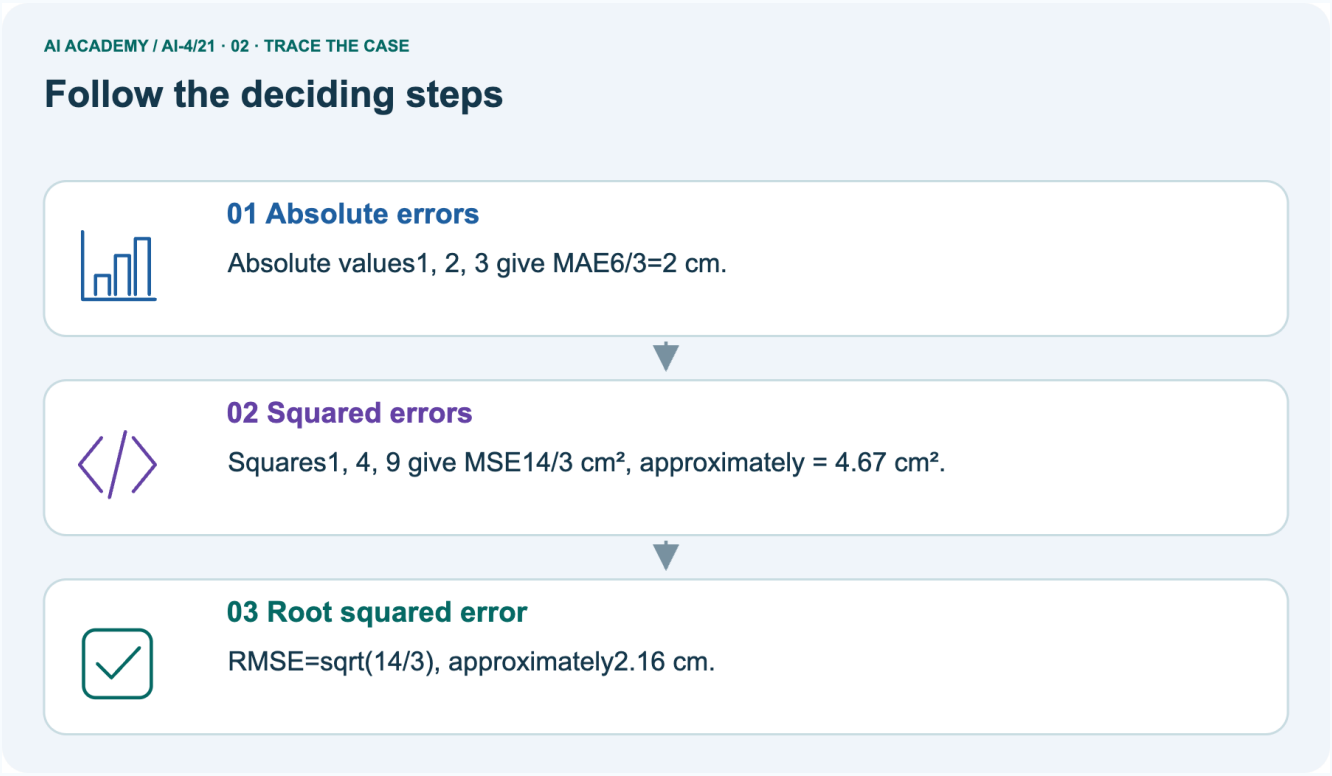
Compare absolute and squared error without losing track of **unit**s.

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main residuals are [-1,2,-3] cm; the pictured residuals are [-1,2,3] cm. Their signs differ but their absolute and squared errors agree: MAE 2 cm, MSE 14/3 cm², RMSE sqrt(14/3)≈2.16 cm. Converting the pictured residuals to mm gives [-10,20,30]: MAE 20 mm, MSE 1400/3 mm², RMSE≈21.60 mm. The physical errors did not change. Never compare the raw cm and mm loss numbers as if the model improved or worsened.

Trace all three losses and mm conversion



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Absolute errors	Absolute values1, 2, 3 give MAE6/3=2 cm.
2. Squared errors	Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm².
3. Root squared error	RMSE=sqrt(14/3), approximately2.16 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Change: Convert every residual to millimetres.

Prediction: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

▼ Additional actual/predicted pair example

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate residuals, **absolute error**s, and mean **absolute error**.

MODEL IDEA **Absolute error** measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17. Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17.

STEP 2 Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

STEP 3 MAE = (2 + 1 + 3) ÷ 3 = 2 cm.

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

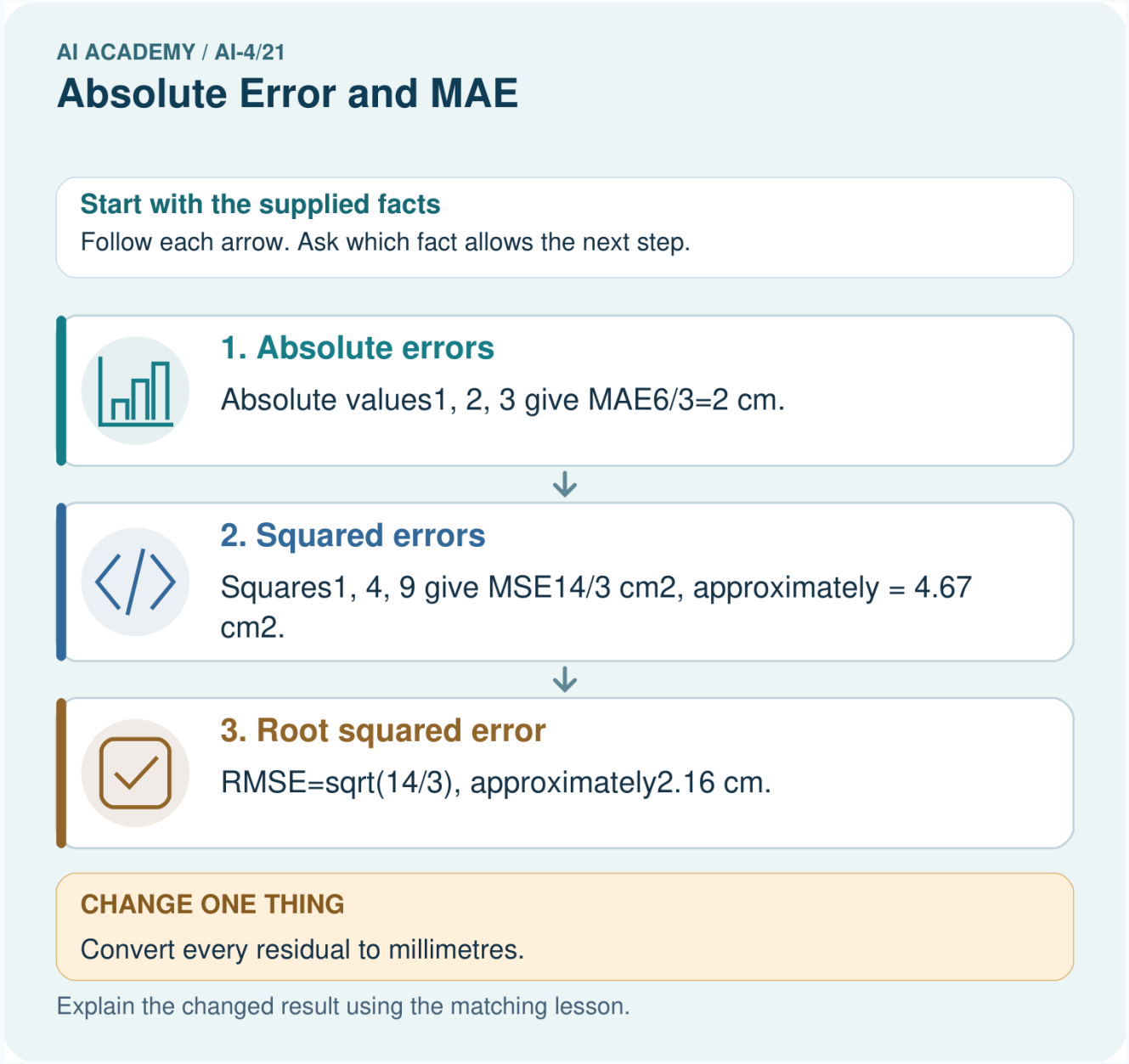
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Absolute Error and MAE

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Give MSE and **RMSE** with **unit**s.

Teacher check: Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Absolute and squared **loss**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Its **unit** is cm², not cm. **RMSE** is the corresponding square-root statistic with cm **unit**s. Changed-case check: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Changed-case reasoning: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Actual=[3,5,9], predicted=[2,7,6] cm.

Write the question to explain: Calculate MAE.

Reveal after **prediction**s: **Absolute error**s=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Use the earlier transfer question as additional practice. Reserve the selected fresh metre-residual and centimetre-conversion task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm. Exit response: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Use the prerequisite question: Which constant minimizes training squared error? Check the explanation before proceeding.
7 to 18 minutes	Read the opening and model the worked evidence. Ask students to identify mae in context.
18 to 32 minutes	Complete W 1 and W 2 with support: compute mae; compute squared loss. Require the evidence, not only the final response.

Stage	Teacher action
32 to 45 minutes	Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.
Session 2 start to 8 minutes	Retrieve the distinction between mae and mse. Invite a counterexample to an overconfident claim.
8 to 20 minutes	Discuss the application and complete W 3: Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?
20 to 35 minutes	Collect W4 independently before giving hints. Assess these outcomes: Calculate MAE MSE and RMSE; Explain how squaring changes sensitivity to large errors. Record any support used.
35 to 45 minutes	Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Main formulas and units as supplied; candidate wins lab MAE2vs3 but loses MSE12vs9 and RMSEsqrt12vs3. Keep the large miss. Zero residuals give zero losses only on those records; missing list has no observed mean. Independent [-2,0,4] m:absolute sum6→MAE2m,squares20→MSE20/3m²,RMSEsqrt(20/3)≈2.582m. In cm:MAE200,MSE200000/3cm²,RMSE≈258.20cm. No final-test or consequence guarantee.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. MAE Lab: calculate each row, verify **unit**s and denominator, then compare model and baseline on identical examples.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

A stronger claim makes a stronger project: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh metre-residual and centimetre-conversion task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average **absolute error**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

Teacher explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Compute MAE

Calculate MAE for the worked residuals.

Teacher explanation: (1+2+3)/3=2 cm.

MAE uses absolute residual sizes, so add 1, 2 and 3 and divide by three: (1+2+3)/3 = 2 cm. Positive and negative signed residuals must not cancel in this calculation.

State centimetres because **absolute error** retains the target **unit**.

W2 Compute squared loss

Give MSE and **RMSE** with **unit**s.

Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

Square the residual sizes to get 1, 4 and 9; their mean is 14/3 square centimetres. Take the square root to obtain **RMSE** sqrt(14/3), approximately 2.16 cm. MSE and **RMSE** describe related quantities but have different **unit**s.

W3 Compare patterns

Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

Teacher explanation: Both absolute sums are 6. Squared sums are 36 and 12, so MSE 12 versus 4.

W5 Choose honestly

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

Teacher explanation: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Additional transfer practice question

Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.

Reasoned practice answer

Absolute errors are 1, 2 and 3 cm. MAE is (1+2+3)/3 = 2 cm. Signed errors must not cancel.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Averaging signed errors is the same as mean **absolute error**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Residuals=[-2,3,-1], summing to 0. **Absolute error**s=[2,3,1], MAE=6/3=2 in the target's **unit**s. Signed errors can cancel even when every **prediction** is wrong.

Explanations for the additional workbook examples

W1 Compute MAE

W2 Compute squared loss

W3 Compare patterns

W4 Apply independently

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

Teacher explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

W5 Choose honestly

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit **unit**s and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use mae and mse in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20. Fresh retry residuals [-1, 1, 4]: MAE 2, MSE 6, **RMSE** sqrt 6 about 2.45 in corresponding **unit**s.

Require **unit**s and operation order. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/22, ai-4/24, ai-4/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 22: Train a Regression Model. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: **Fitting** a line. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Calculate MAE MSE and **RMSE**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Give MSE and **RMSE** with **unit**s.

Reasoned practice answer: Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

Check every learner before discussion: Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Convert every residual to millimetres. Require a **prediction** and a reason before comparing. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

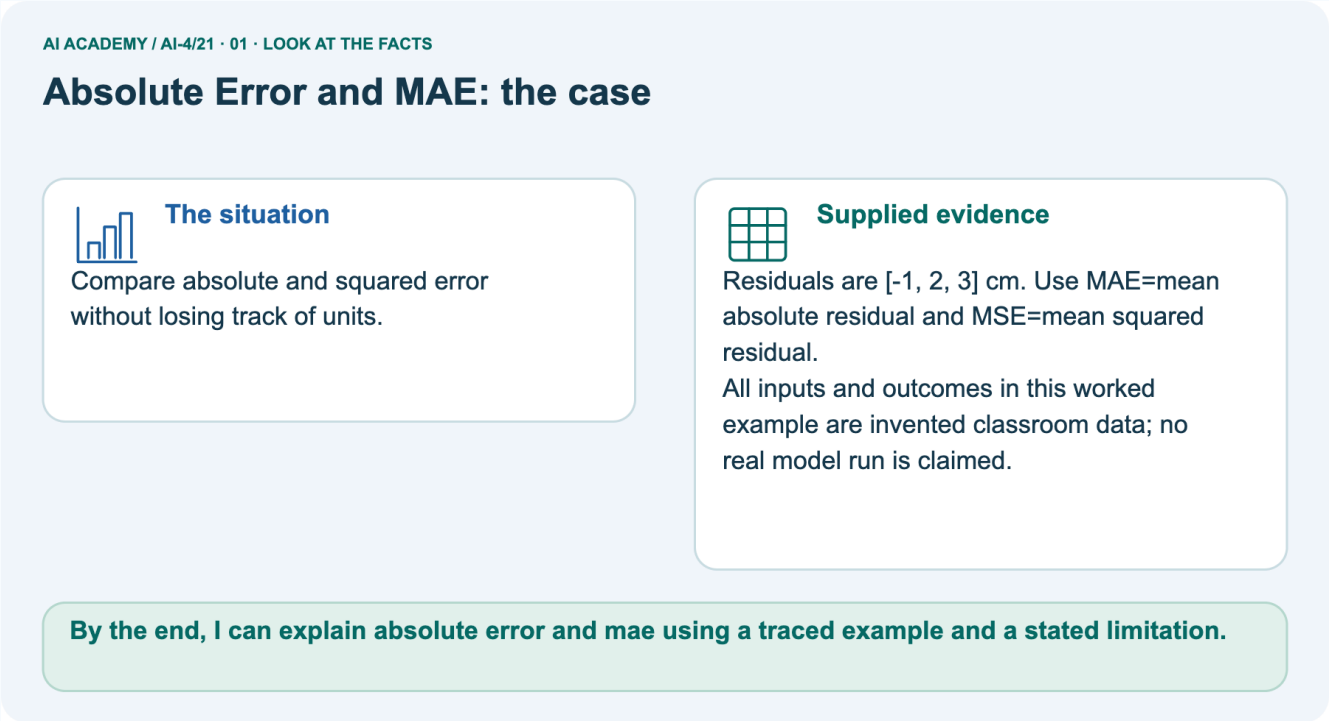
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Absolute Error and MAE: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

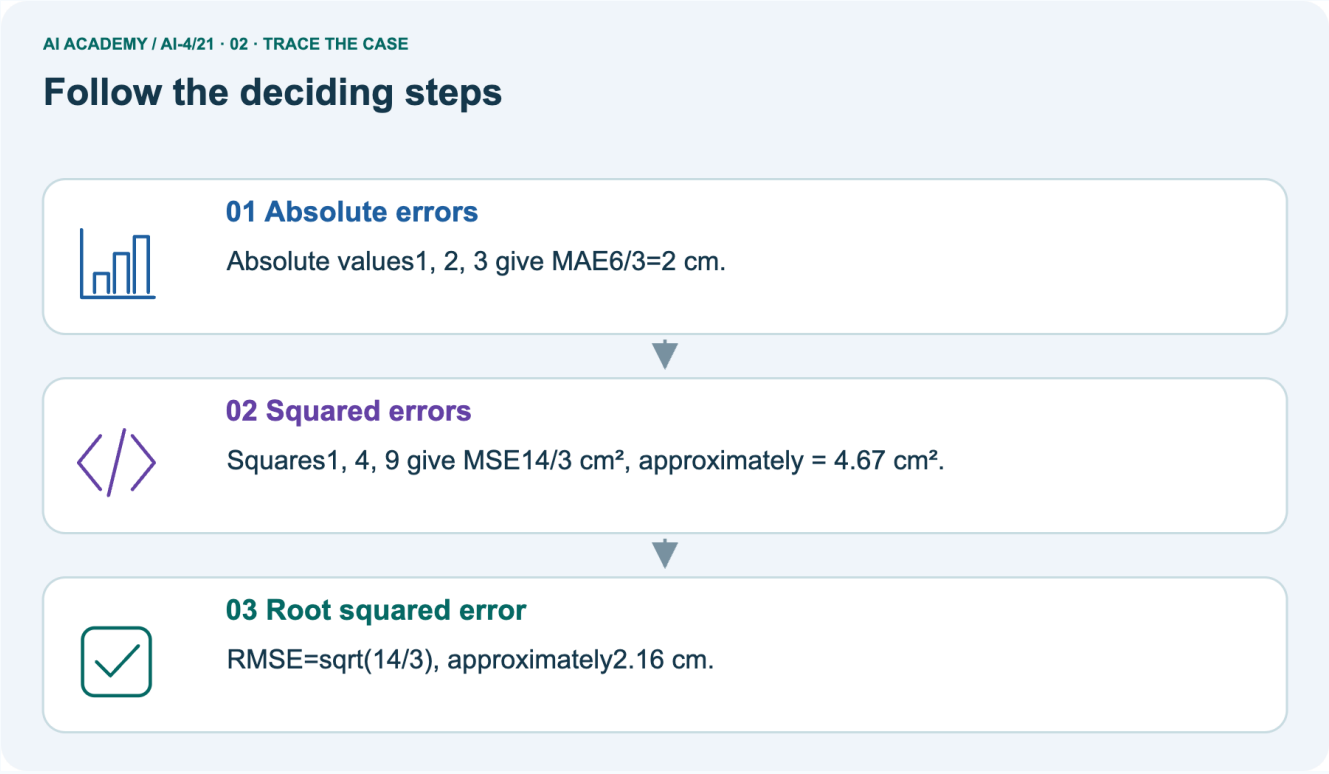
Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



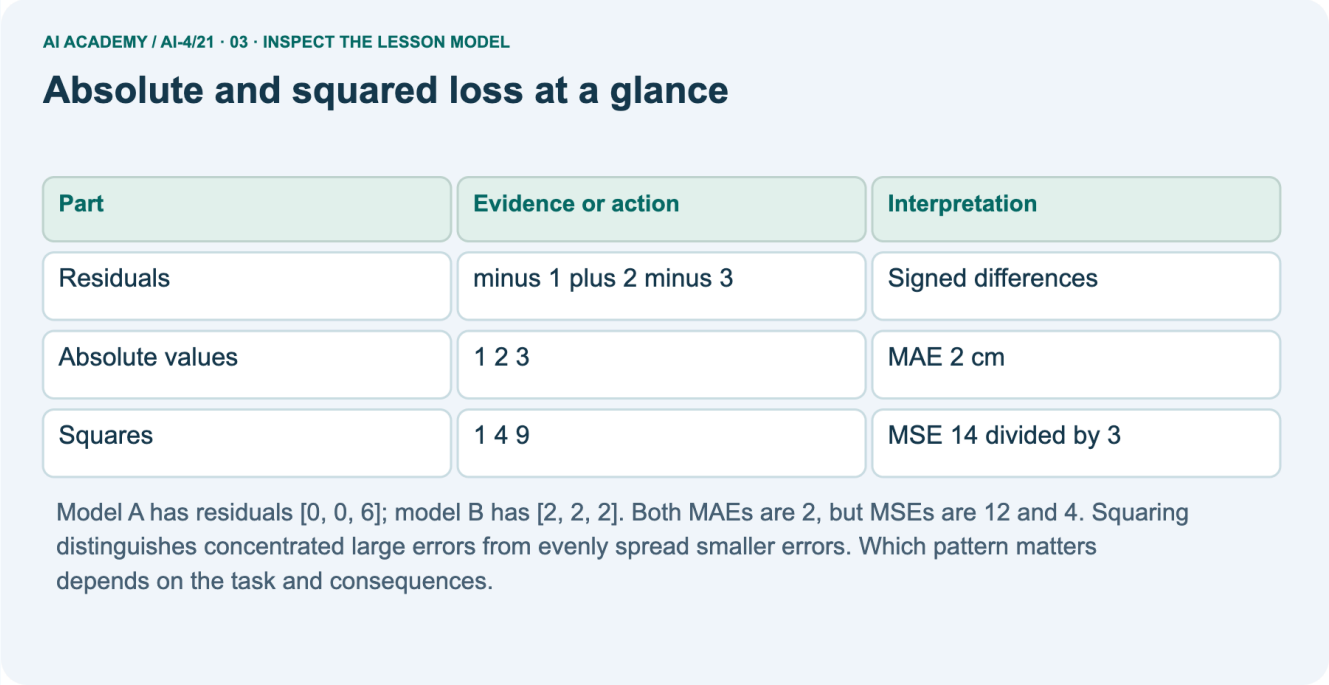
Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute errors: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Absolute and squared loss at a glance



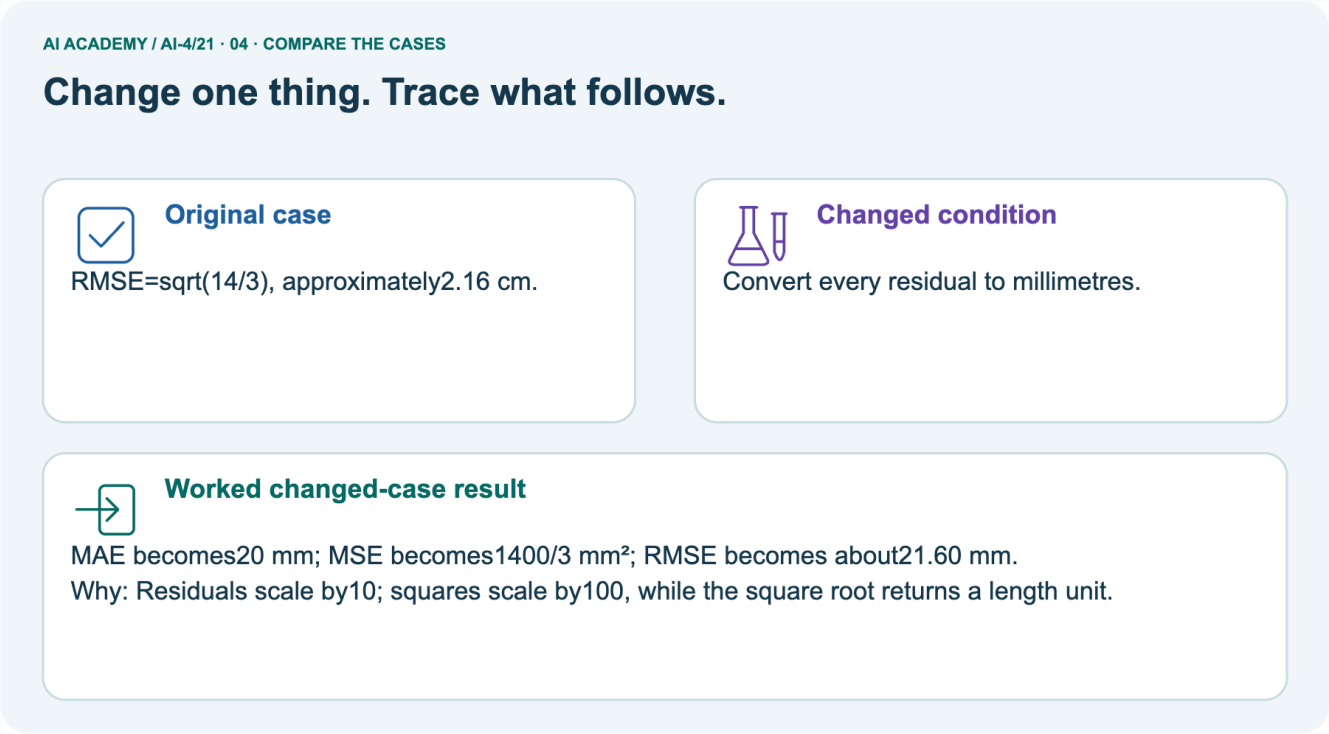
Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

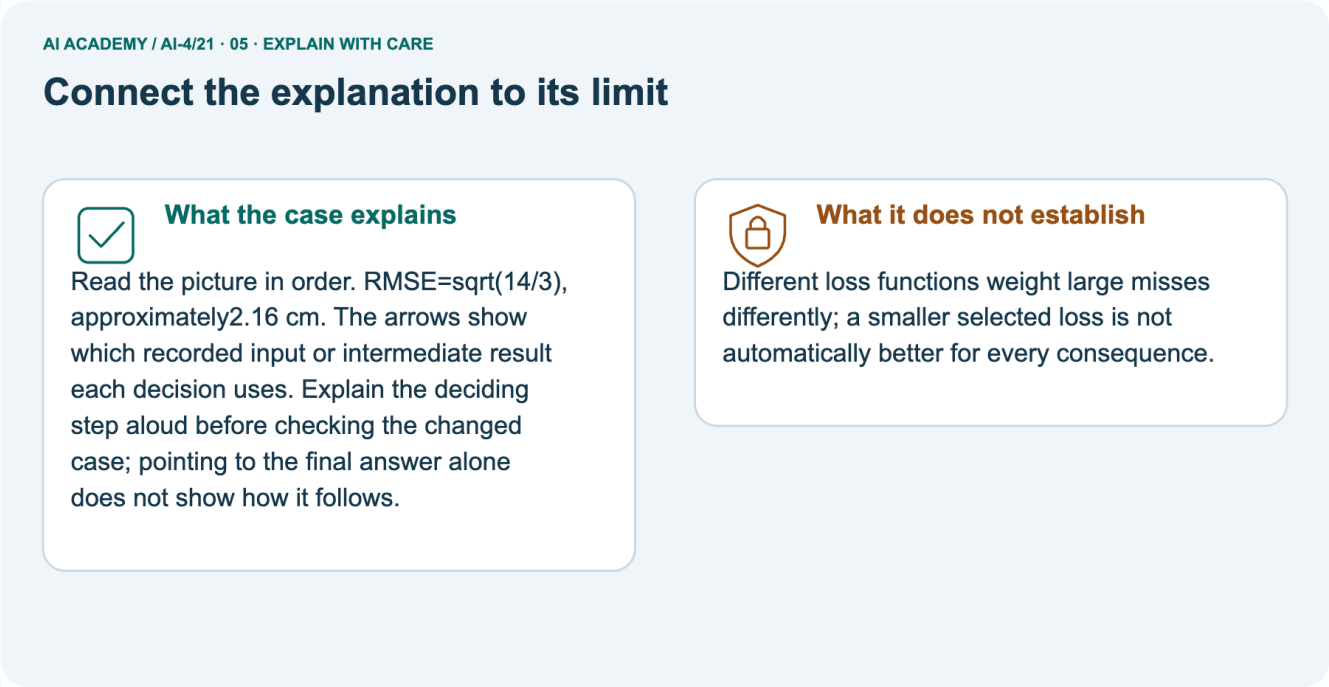
Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

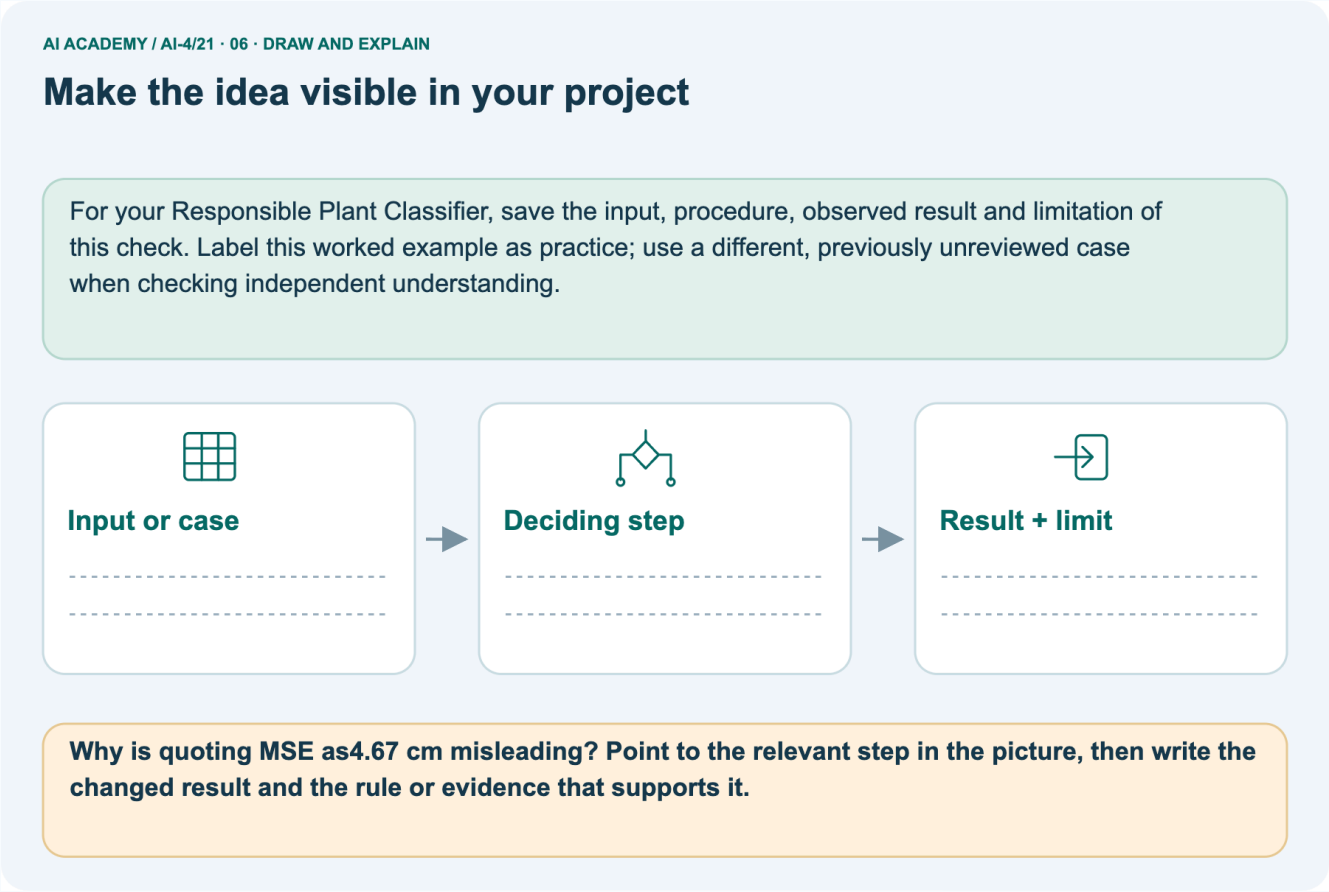
Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Its **unit** is cm², not cm. **RMSE** is the corresponding square-root statistic with cm **unit**s. Changed-case check: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Absolute Error and MAE: the case

02. Follow the deciding steps

03. Absolute and squared loss at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.